

Smart Parking Platform: Product Backlog

Login

1. As a driver, I want to create an account with my email or phone number so I can book parking spots.
2. As a driver, I want to log in with saved credentials so I can access my account quickly.
3. As a user, I want to reset my password via email if I forget it so I can regain access to my account.
4. As an operator, I want a separate role-based login so I can access operator-only tools and dashboards.
5. As a system admin, I want to manage user roles and permissions so I can control access to sensitive backend systems.
6. As a driver, I want to stay logged in on trusted devices so I don't have to re-enter credentials every time.
7. As a user, I want optional two-factor authentication so my payment info stays protected.

UI for Operator and User

Driver-facing

8. Home screen with a map view showing nearby available spots.
9. Spot search screen with filters (price, distance, covered/uncovered).
10. Reservation screen with time slot and duration selection.
11. Payment screen with saved card options.
12. Booking confirmation screen with QR code or entry code.
13. Booking history and digital receipts screen.
14. Profile and account settings screen.
15. Notifications screen for upcoming or expiring reservations.
16. Dispute/issue reporting screen.

Operator-facing

17. Operator dashboard showing real-time occupancy across their lot(s).
18. Spot/zone management screen to add, remove, or edit parking spots.
19. Pricing control screen to set fixed or dynamic rates.
20. Dispute resolution screen for reviewing driver complaints.
21. Reservation management screen showing current and upcoming bookings.
22. Staff access management screen for lot employees.
23. Lot performance summary widget on the dashboard.

Backend Processes for Operator and User

24. Reserve a parking spot and lock it from other bookings.
25. Release a reserved spot if payment isn't completed within a time window.

26. Process payments and handle transactions.
27. Process refunds for canceled reservations.
28. Sync real-time spot availability across all connected devices.
29. Dispatch notifications (email/SMS/push) for confirmations and reminders.
30. Create and route dispute tickets to the relevant operator.
31. Log all reservation and payment activity for auditing.
32. Update operator inventory when spots are added or removed.
33. Calculate dynamic pricing based on demand and time of day.
34. Manage user authentication and sessions on the backend.
35. Validate booking requests to catch time conflicts.
36. Automatically expire and clean up unpaid or abandoned reservations.
37. Integrate with a third-party payment processor via API.
38. Reconcile daily transactions for each operator.
39. Flag suspicious booking patterns for fraud review.

Reporting for Operator and User

40. Driver: parking history report of past reservations.
41. Driver: monthly spending summary.
42. Driver: downloadable/exportable receipts.
43. Operator: daily revenue report by lot.
44. Operator: occupancy rate report by lot and time of day.
45. Operator: dispute log and resolution report.
46. Operator: peak usage hours report.
47. Admin: platform-wide revenue and usage report across all operators.
48. Admin: user growth and retention report.
49. Operator: no-show and cancellation rate report.
50. Driver: time-saved report from reduced parking search time.

